

EL403 – Assignment 3

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About the document — In this document, we have presented our understanding, analysis and conclusions on the given exercise- the Design of Digital FIR from the given information and for the specifications.

Given Information:

Filter Specification:

- Filter Type: Low-Pass Filter
- Cut-off frequency at 6 dB = 1kHz
- Stop-band attenuation of 20 dB = 1.1 kHz
- Number of Taps (N) = upto 15

Coefficients:

b = [-0.04557, 0, 0.06366, 0, -0.1061, 0, 0.3183, 0.5, 0.3183, 0, -0.1061, 0, 0.06366, 0, -0.04557]

Input samples and Coefficients are of 8-bit.

Determination of sampling frequency

We have determined the sampling frequency (fs) at which the sinc function was sampled using the given specification to be 4000 Hz.

Choice of Representation format for Input samples and filter coefficients

Coefficients:

Since,

- all the given coefficients are within the range of $[-1, 0.99218]$,
 - 8 bits are available and,
 - the previously designed adders and multipliers support 2's complement operations,
- we chose to represent them in Q1.7 format i.e., 1 sign bit and 7 fractional

Input Samples:

We have chosen to represent the input samples also in Q1.7 format. There will not be any problem until the sample value within the range, $[-1.0.99218]$.

Note:

Depending upon the application, if significant number of input values lie outside the range of Q1.7 numbers, to avoid saturation of the input values, the input can be normalized back to the acceptable range by down scaling it by a factor of

$$\frac{A_{max}}{1 - 2^{-f}}$$

Where,

A_{max} the maximum input value possible for that application, f is the number of fractional bits available.

After the computation, multiplying each output value by the same factor will give scaled the output.

Effects on Magnitude and Phase of Frequency Response due Conversion of coefficients to Q 1.7

The given coefficients are rounded to nearest values that can be represented accurately using Q1.7 format are as follows:

Code:

```
bq = fi(b,1,8,7,'RoundingMethod','Nearest');
```

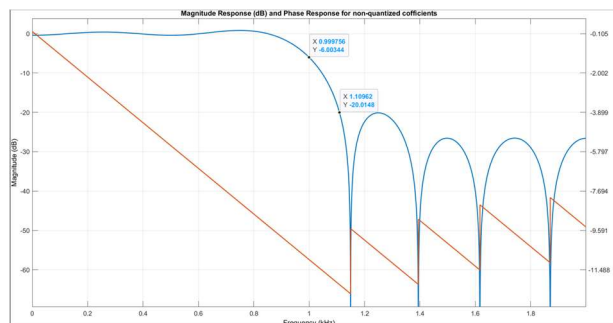
bq = [-0.046875, 0, 0.0625, -0.109375, 0.3203125, 0.5, 0.3203125, 0, -0.109375, 0.0625, -0.046875];

bq is the vector of *quantized coefficients*.

The following are the magnitude and phase responses of the filter before and after coefficient quantization:

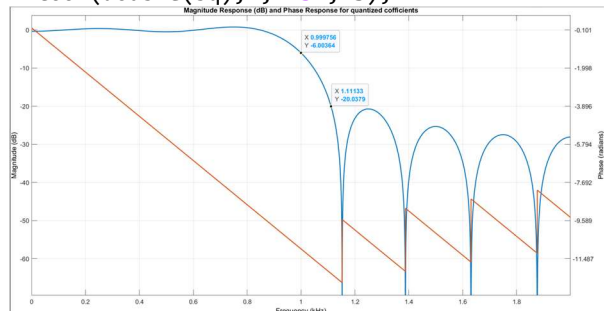
a. Before Quantization:

```
fvtool(double(b),1,'Fs',fs);
```



b. After Quantization:

```
fvtool(double(bq),1,'Fs',fs);
```



Observations:

- i. Since the coefficients show a positive symmetry, the FIR will have a constant phase delay and group delay of

$$d_n = \frac{\text{slope}}{2\pi} \times F_s = \frac{N-1}{2} \text{ samples}$$

with respect to the input. Note that the variations in the phase response on the plot is only because of the numerical approximations in MATLAB.

Here, slope = 0.011 and N = 15 gives,

$$0.011/(2 \times 3.14) \approx 7 = (15-1)/2.$$

- ii. Change in -6 dB frequency and stopband attenuation,

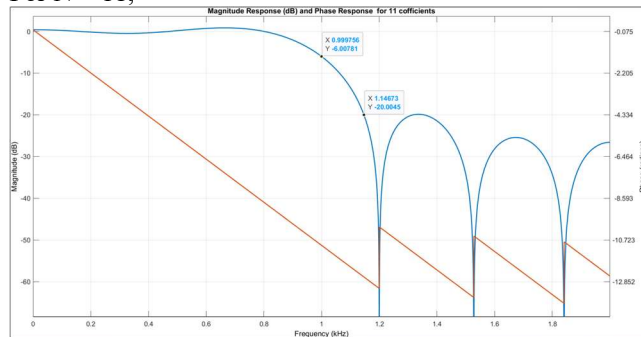
Δ in -6dB frequency is almost zero.

$$\Delta \text{ in } -20 \text{ dB frequency} = \frac{1.11133 - 1.10962}{1.10962} \times 100 = +0.154 \%$$

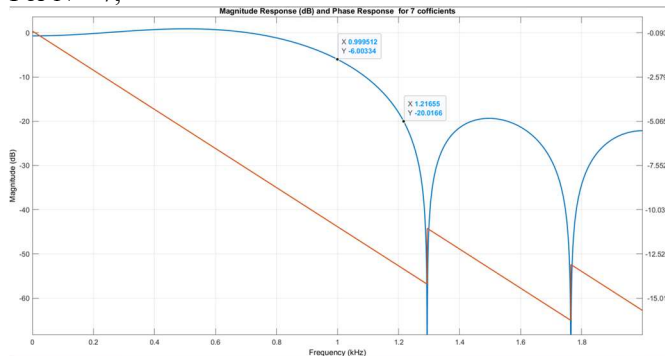
Though there is no change in -6 dB point, transition width of the filter increases. Also, stop band attenuation increase by approx. 0.48 % in dB.

Effect on the Frequency Response when the number the number of coefficients are reduced

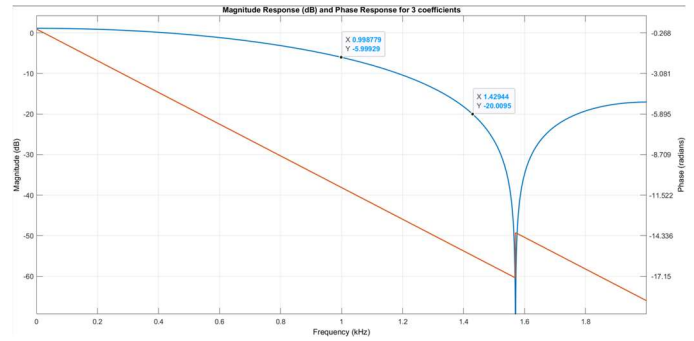
For N = 11,



For N = 7,



For N = 3,



Observations from the plots:

Reducing the number of taps,

1. significantly increases the transition width. (roll-over becomes steeper)
2. phase delay reduces since is equal to (N-1)/2 and,
3. there is no change in -6 dB frequency.

Effects of Finite Word Length on Filter Output

Assumption: We have assumed a sinusoidal input with a low amplitude sinusoidal noise as our input for the below analysis and the decision making.

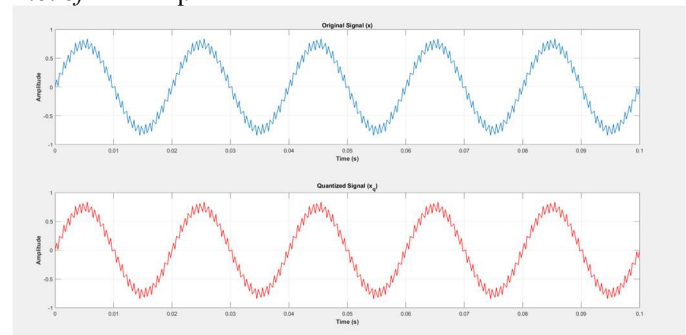
$$x = 0.75 \sin(2\pi \cdot 50 \cdot t) + 0.1 \sin(2\pi \cdot 1500 \cdot t);$$

$$xq = \text{fi}(x, 1, 8, 7,);$$

$$\text{snr_value} = \text{snr}(\text{double}(xq), \text{double}(x - \text{double}(xq)));$$

where, x is unquantized input and xq is quantized to 8-bit (Q1.7).

Plot of x and xq:

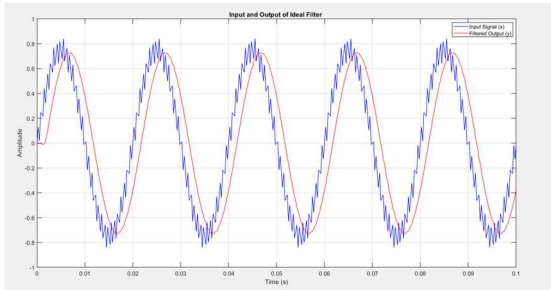
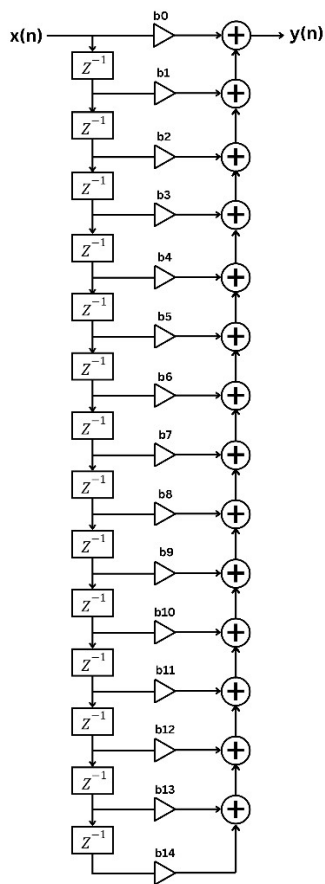


Quantization noise in the input signal:

$$\text{SQNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{qua noise}}} \right) = 20 \log_{10} \left(\frac{xq}{x-xq} \right) = 41.72 \text{ dB}$$

Power Interpretation:

$$P_{\text{signal}} = 10^{4.172} P_{\text{quant noise}} \approx 14859 P_{\text{quant noise}}$$

For an ideal filter:(Finite Word-length effects are *not* considered) $y = \text{filter}(b, 1, x);$ *Input and Output Plot:***For Direct Form realization (with Finite Word length):**

Resources needed:

- 15 Multipliers
- 14 Adders
- 14 Delay Elements
- 15 Memory Elements

*Fig-1 : Direct Form Architecture.**MATLAB Model:*

```

y = fi(zeros(size(xq)), 1, 8, 7, 'RoundingMethod',
'Floor');
N = 15;
for n = N:length(xq)
    acc = fi(0, 1, 8, 7, 'RoundingMethod', 'Floor');
    for k = 1:N
        mult = bq(k) * xq(n - k + 1);
        acc = acc +
fi(mult, 1, 8, 7, 'RoundingMethod', 'Floor');

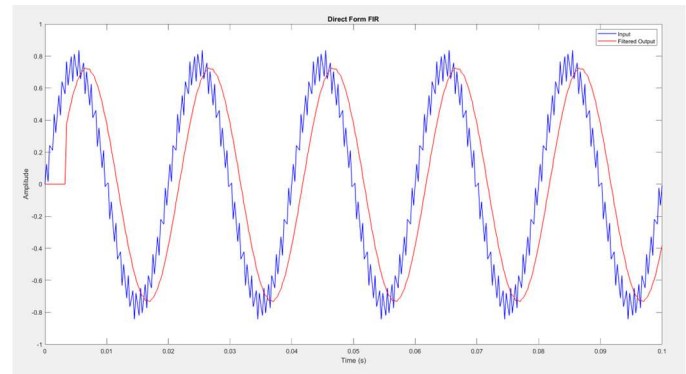
```

```

acc =
fi(acc, 1, 8, 7, 'RoundingMethod', 'Floor');
end
y(n) = acc;
y(n) =
fi(y(n), 1, 8, 7, 'RoundingMethod', 'Floor');

```

end

Input and Output Plot:

$$SQNR = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{quant noise}}} \right) = 20 \log_{10} \left(\frac{yq}{y-yq} \right) = 26.03 \text{ dB}$$

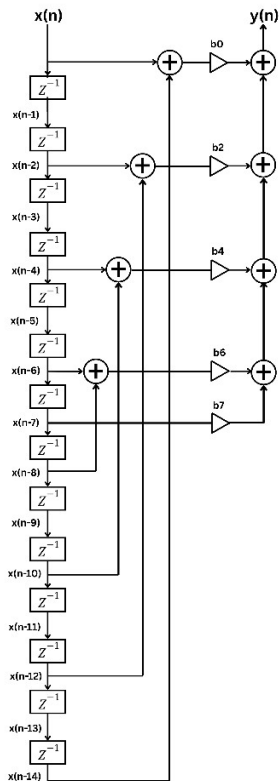
Power Interpretation:

$$P_{\text{signal}} = 10^{2.603} P_{\text{quant noise}} \approx 401 P_{\text{quant noise}}$$

For Optimized structure:

We have optimized the Direct Form structure to the following structure exploiting the following facts:

1. Alternate zeros in the coefficients makes the number of multipliers need half.
2. Positive Symmetry: The samples that are to be multiplied with the same coefficients can be grouped and pre-added before multiplying with the coefficients, again halving the multiplier requirement.



Resources needed:

- 5 Multipliers
- 8 Adders
- 14 Delay elements
- 5 Memory Elements

Fig-2. Optimized Architecture

MATLAB Model:

```
y_opti = fi(zeros(size(xq)),1,8,7,
'RoundingMethod','Floor');
N = length(bq);
```

```
for n = N:length(xq)
    s0 = xq(n) + xq(n-14);
    s1 = xq(n-2) + xq(n-12);
    s2 = xq(n-4) + xq(n-10);
    s3 = xq(n-6) + xq(n-8);
    xc = xq(n-7);

    p0 = bq(1)*s0; p1 = bq(3)*s1;
    p2 = bq(5)*s2; p3 = bq(7)*s3;
    pc = bq(8)*xc;
```

```
acc =
fi(p0+p1+p2+p3+pc,1,8,7'RoundingMethod','Floor');
y_opti(n) = acc;
```

```
end
```

$$SQNR = 10 \log_{10} \left(\frac{P_{signal}}{P_{quant\ noise}} \right) = 20 \log_{10} \left(\frac{y_{opti}}{y-y_{opti}} \right) = 37.95 \text{ dB}$$

Power Interpretation:

$$P_{signal} = 10^{3.795} P_{quant\ noise} \approx 6237 P_{quant\ noise}$$

Strengths of the Reduced structure:

Far less resources and improved SQNR as compared to direct form implementation.

Word-length decisions:

- 8-bit adders can be used for pre-additions and truncate the sum to 8-bit.
- 8-bit multipliers for multiplying the sums and truncate the product to 8-bit.
- 8-bit Multi-operand adders for adding the products and truncate the accumulator to 8-bit.

Trade-offs for the optimized structure:

There is no chance overflow at the pre-addition stage for the assumed input signal but, depending on the application overflow may occur and could be avoided by,

- i. Scaling down the input samples and/or coefficients by a suitable scaling factor. (As explained in Pg. 1)
- ii. Having guard bits to represent the sum and round-off.

Observation:

The given set of coefficients have alternate zeroes and 0.5 for the centre coefficient, it gives a **half-band filter**.

Advantages of half-band filters:

1. Number of multipliers required would be halved.
2. The cut-off frequency is quarter the sampling frequency (i.e., the half the Nyquist sampling frequency). Hence, it removes all the frequency components in the upper half band. **So, the output can sampled at half the original sampling frequency.** That implies, a slower frequency DAC (half the sampling rate of ADC) would be sufficient for reconstructing the output.